
BRAVO COMPANY
COMMUNICATIONS DOMINATE WARFARE

DATA SYSTEMS ADMINISTRATOR COURSE

DS.02.02

SERVER ROLES AND CAPABILITIES

OVERVIEW

- Server Roles
- RSAT
- ADAC
- Metadata Cleanup
- REPADMIN Tools

LEARNING OBJECTIVES

- Refer to your student handout for detailed list of your Terminal and Enabling learning objectives.

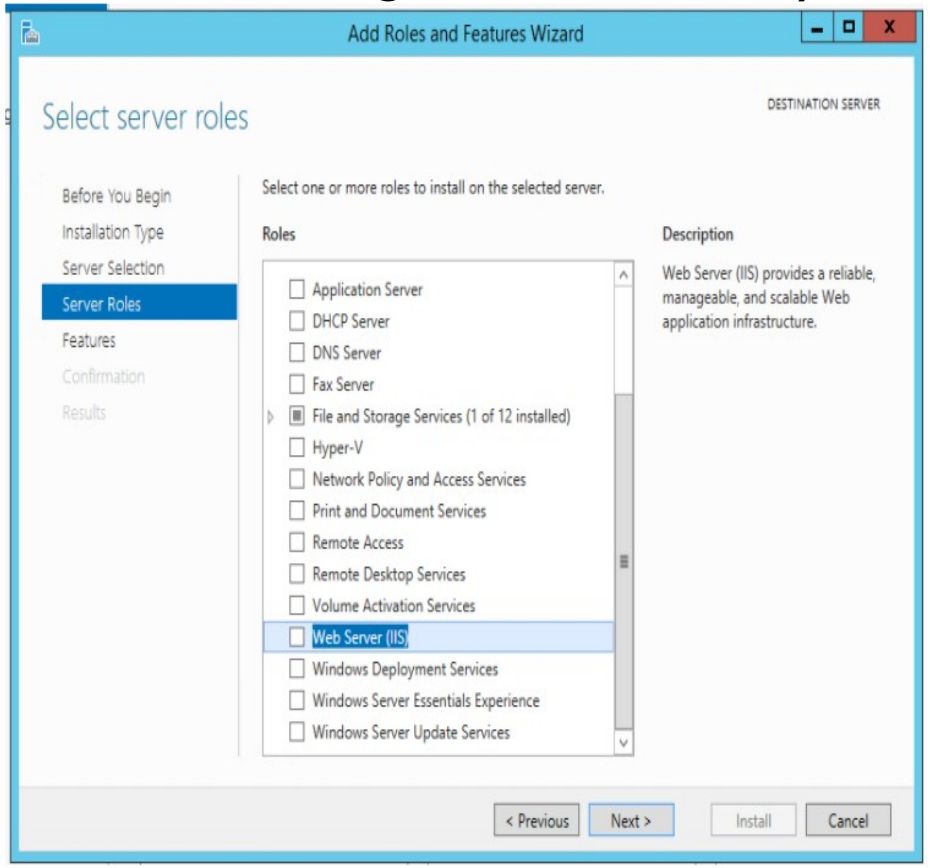
SERVER ROLES

SERVER ROLES

- ***What is a Server Role?***
  - A server role is a set of software programs that, when they are installed and properly configured, lets a server perform a specific function for multiple users or other computers within a network
  - They describe the primary function, purpose, or use of a server.
    - DC = Domain Controller – Controls the Domain (covered later in this class)

SERVER ROLES

- ***How many roles can a server have?***
  - A server can have at most 250 roles. (Usually separated among other servers)
  - Most servers may have up to three different roles
    - *Common on many USMC Networks*
  - This is all planned out during the initial configurations and implementation



SERVER ROLES

- ***Active Directory Domain Services (ADDS)***
  - A role in Active Directory that allows admins to manage and store information about resources
  - AD DS helps admins manage both computing devices and users
  - Will reorder them into a custom hierarchical structure
  - AD DS also integrates security by authenticating logons and controlling access to directory resources
  - All types of USMC networks utilize AD DS in some fashion or form whether on tactical or in garrison.

SERVER ROLES

- ***Active Directory Domain Services (ADDS)***
  - ***How is Active Directory Domain Services used?***
    - It is used for identity and access management
    - Stores and organizes information about the people, devices and services connected to a network.
    - A central point of administration for all activity on the organizationsa network.
    - Can be used on premise or in hybrid cloud environments

SERVER ROLES

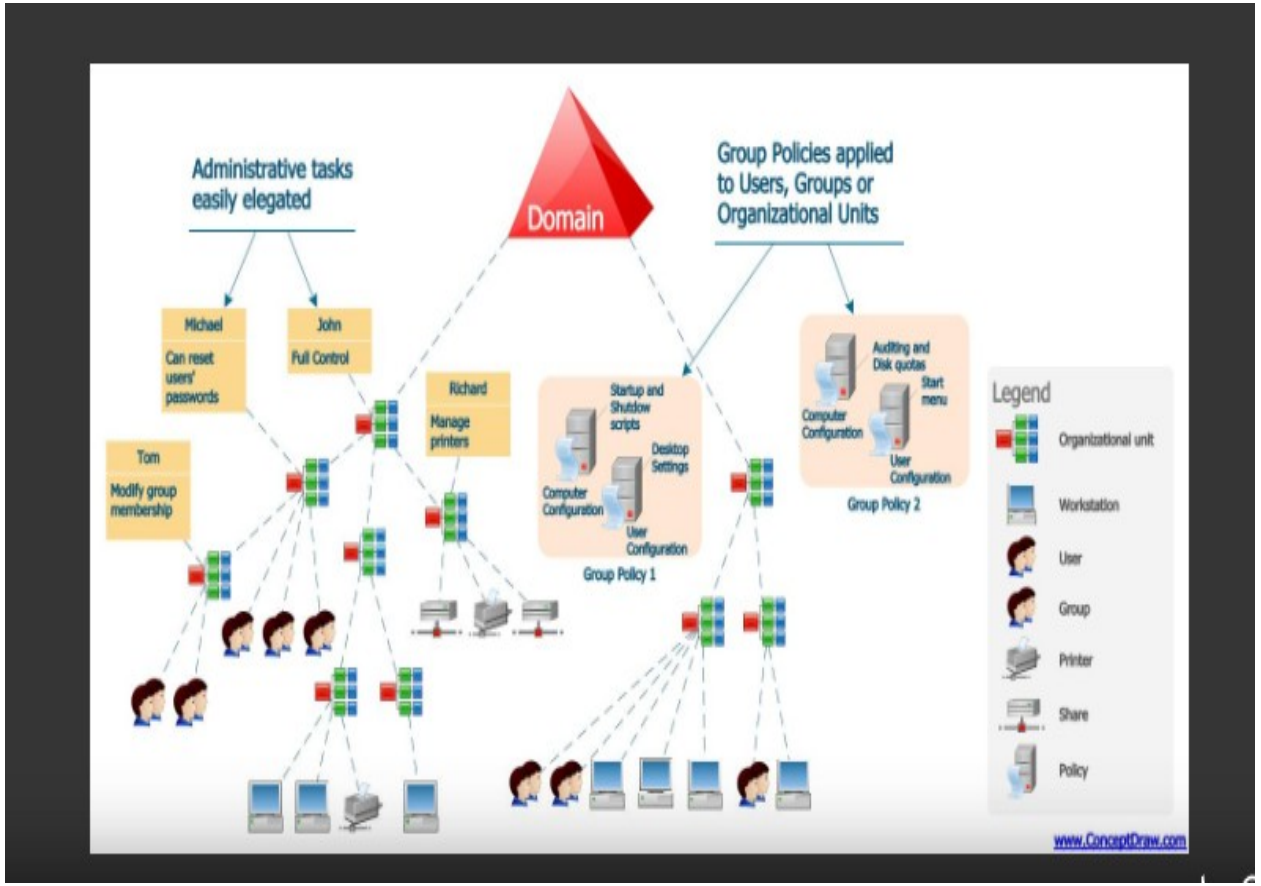
- ***Active Directory Domain Services (ADDS)***
  - ***How does AD DS work?***
    - Lets various Domain Services connect with them and users access or manage them
    - The hierarchical structure includes the following:
      - **Domains**
      - **Organizational units**
      - **Active Directory trees**
      - **Active Directory forests**

SERVER ROLES

- ***Active Directory Domain Services (ADDS)***
  - ***What are the benefits of Active Directory Domain Services?***
    - Hierarchical structure
      - This is the main benefit of AD DS
    - Flexibility
      - Users can access Active Directory from any computer on the network.
    - Single point of access
      - Creates a single point of access to network resources
    - Redundancy
      - Has built in replication and redundancy

SERVER ROLES

- ***Active Directory Domain Services (ADDS)***



SERVER ROLES

- ***Active Directory Sites and Services***
  - The information in Sites & Services is used by AD to create a replication topology, determine which DCs should authenticate clients, and help users connect to applications and services
  - Misconfigurations can result in increased logon times, poor user experience with some applications.
  - A directory service site topology is a logical representation of the physical network

SERVER ROLES

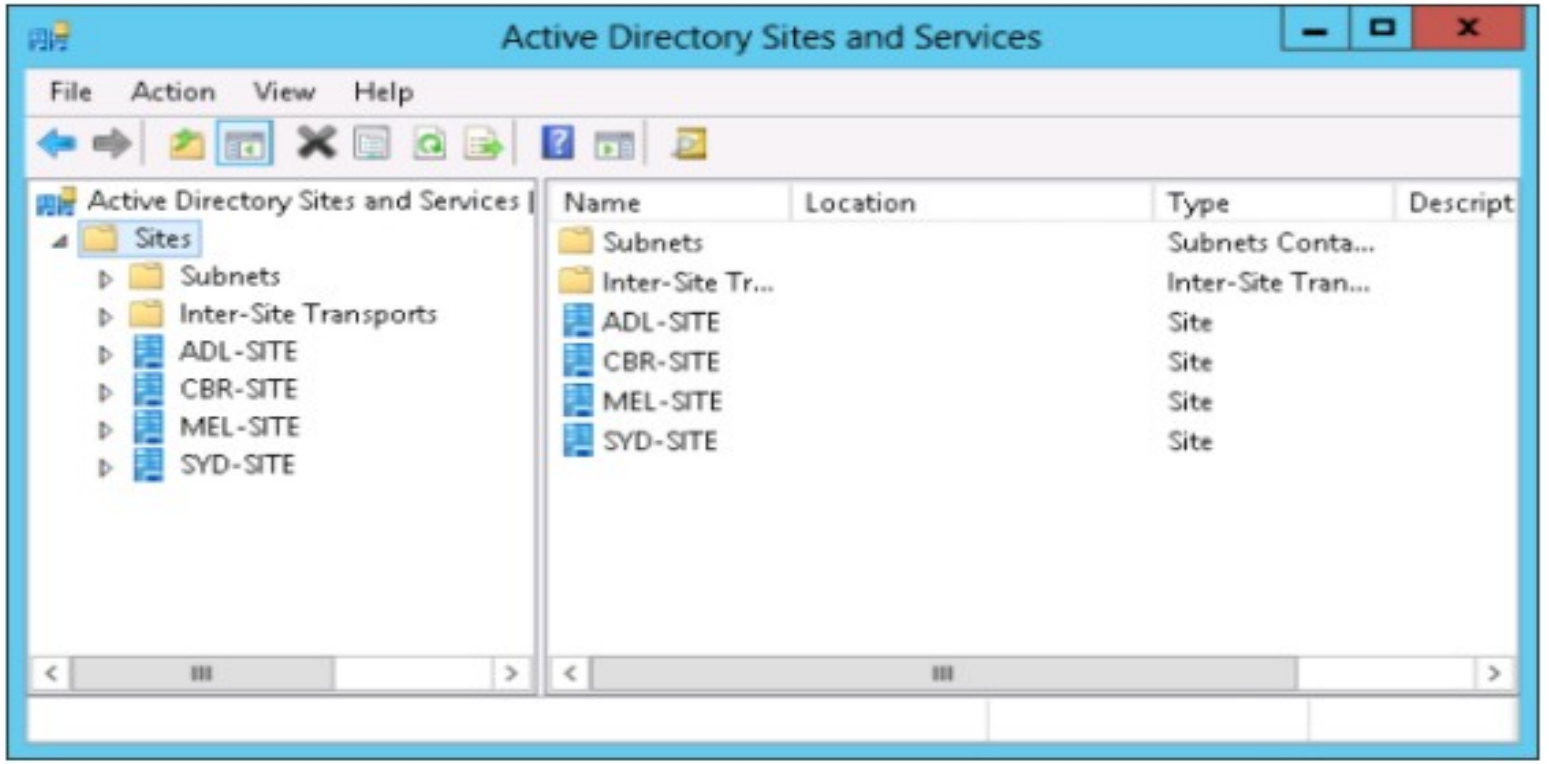
- ***Active Directory Sites and Services***
  - Involves planning for domain controller placement and designing sites, subnets, site links, and site link bridges to ensure efficient routing of query and replication traffic.
  - Helps administrators efficiently route client queries and Active Directory replication traffic

SERVER ROLES

- ***Active Directory Sites and Services***
  - A well-designed site topology helps any organization achieve the following benefits:
    - Minimize the cost of replicating Active Directory data.
    - Minimize administrative efforts that are required to maintain the site topology.
    - Schedule replication during off-peak hours.
    - Helps to reduce network traffic over slow wide area network (WAN) links, improve logon and logoff processes, and speed up file download operations.

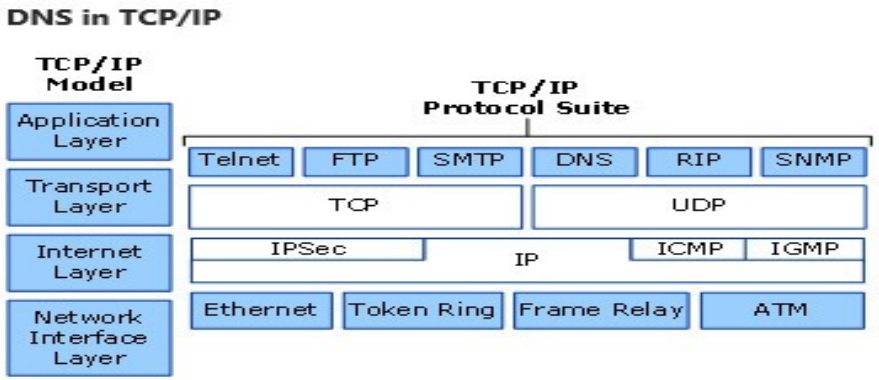
SERVER ROLES

- ***Active Directory Sites and Services***



SERVER ROLES

- ***Domain Name System (DNS)***
  - Provides computer name-to-IP address mapping name resolution services to computers and users
  - Computers use DNS to locate Active Directory domain controllers
  - Domain controllers use DNS to locate each other



SERVER ROLES

- ***Domain Name System (DNS)***
  - ***Benefits of DNS?***
    - **Application Load Balancing**
      - Administrators can use DNS policy to balance the traffic load between the different application instances
    - **Split Brain DNS**
      - DNS records are split into different Zone Scopes on the same DNS server
    - **Forensics**
      - Administrators can use DNS policy to redirect malicious DNS clients to a non-existent IP address instead of directing them to the computer they are trying to reach

SERVER ROLES

- ***Domain Name System (DNS)***
  - ***Types of DNS Records***
    - **Static record**
      - Can only be updated or deleted from the DNS server on-command by the administrator or another program
    - **Dynamic record**
      - Usually created automatically by a client machine that obtains IP address from a DHCP server.
      - The timestamp indicates the record validity period
    - **Host (A) record**
      - Most basic type of record that exists in the DNS

SERVER ROLES

- ***Domain Name System (DNS)***
  - ***Types of DNS Records (Continued)***
    - **Host (A) record**
      - It maps a hostname string into its IP address
      - Can only be created and stored in a forward lookup zone
      - Can be created as a static or dynamic record.
      - Every Host A-record should have a corresponding **Pointer record** in the reverse lookup zone.
    - **AAAA record (Quad A)**
      - AAAA record is using IPv6

SERVER ROLES

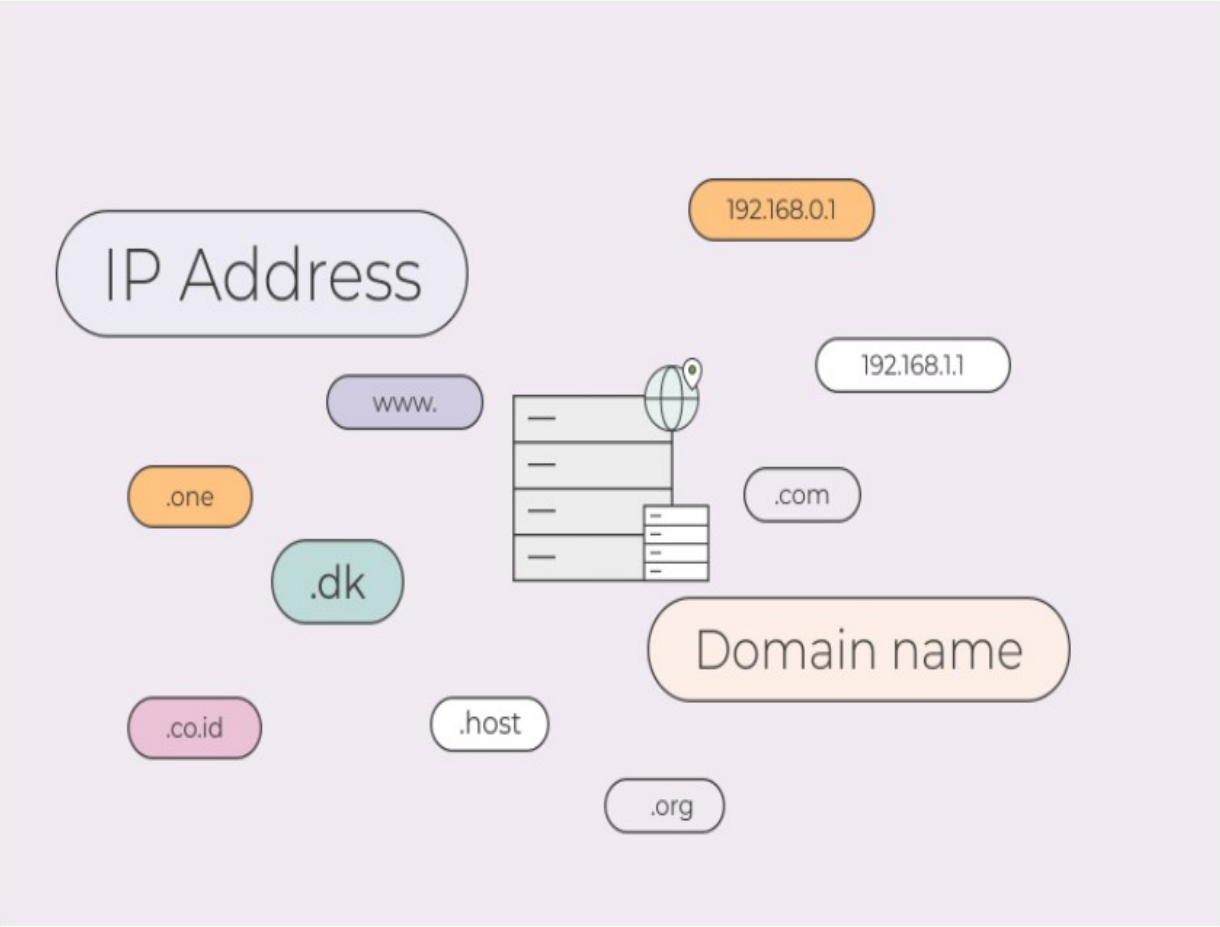
- ***Domain Name System (DNS)***
  - ***Types of DNS Records (Continued)***
    - **Pointer (PTR) record**
      - Pointer or PTR record is the counterpart of an A-record
      - The PTR record can only be stored in a **reverse lookup zone**
      - If the Reverse lookup zone is not created then the Pointer records will not be automatically created
    - **Alias (CNAME) record**
      - The mapping of a hostname into another hostname
      - Multiple CNAME can be created pointing to the same A record
      - CNAME can be created in a different zone name than the A record.

SERVER ROLES

- ***Domain Name System (DNS)***
  - ***Types of DNS Records (Continued)***
    - **Mail Exchanger (MX) record**
      - Used specifically for the purpose of sending email
      - Indicates that the domain is willing to receive email
      - Usually one MX record per “Exchange Server”
    - **Service (SRV) record**
      - Used for service discovery
      - Contains several information such as the priority, weight, service port number, and the host name where the service is residing
      - Chat Solutions like Cisco Jabber rely on SRV records for DNS resolution.

SERVER ROLES

- ***Domain Name System (DNS)***



SERVER ROLES

- ***Domain Controller Roles***
  - Has the Active Directory Domain Services (ADDS) role installed
  - By default, a domain controller stores one domain directory partition consisting of information about the domain in which it is located
  - Specialized domain controller roles that perform specific functions in an ADDS environment

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SERVER ROLES

- ***Domain Controller Roles***
- **Flexible Single Master Operations Master (FSMO) roles (5)**
  - **Schema Master**
  - **Domain Naming Master**
  - **Relative Identifier (RID) Master**
  - **Primary Domain Controller**
  - **Infrastructure Master.**

SERVER ROLES

- ***Domain Controller Roles***
- **Flexible Single Master Operations Master (FSMO) roles (5)**
  - **Schema Master**
    - 1 Per Forest
    - Responsible for processing updates to AD schema
    - Only one that can process updates to the directory schema
    - Typically administered by a NCO, SNCO or even Chief Warrant Officer.

SERVER ROLES

- ***Domain Controller Roles***
- **Flexible Single Master Operations Master (FSMO) roles (5)**
  - **Domain Naming Master**
    - 1 Per Forest
    - Responsible for managing the addition and removal of domains and application partitions in the forest
    - Only one that can add or remove a domain from the directory

SERVER ROLES

- ***Domain Controller Roles***
- **Flexible Single Master Operations Master (FSMO) roles (5)**
  - **Primary Domain Controller (PDC)**
    - 1 Per Domain
    - Responsible for replicating password changes
    - Also performs time synchronization across domain members and is responsible for managing GPOs
    - Also Account lockouts, and Authentication failures

SERVER ROLES

- ***Domain Controller Roles***
- **Flexible Single Master Operations Master (FSMO) roles (5)**
  - **Infrastructure Master**
    - 1 Per Domain
    - Keeps track of changes made in other domains in the forest and their impact on objects in the local domain

SERVER ROLES

- ***Domain Controller Roles***
- **Flexible Single Master Operations Master (FSMO) roles (5)**
  - **Infrastructure Master**
    - 1 Per Domain
    - Keeps track of changes made in other domains in the forest and their impact on objects in the local domain
    - Responsible for updating the group-to-user references whenever the members of groups are renamed or changed

SERVER ROLES

- ***Domain Controller Roles***
- **Flexible Single Master Operations Master (FSMO) roles (5)**
  - **Relative Identifier Master (RID)**
    - 1 Per Domain
    - Responsible for processing RID Pool requests from all DCs within a given domain
    - Controls the RID pool

SERVER ROLES

- ***Active Directory Federation Services (ADFS)***
- ***What is ADFS?***
  - A Single Sign-On (SSO) solution created by Microsoft.
  - Provides users with authenticated access to applications
  - Simplifies the user experience
  - Uses a Federated Trust, linking ADFS and the target application to grant access to users
  - ****Good examples of applications that are ADFS integrated are; Cisco Jabber, Solarwinds, Web Help Desk, SharePoint, and even Transverse Chat.***

SERVER ROLES

- ***Dynamic Host Configuration Protocol (DHCP)***
  - Client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address
  - DHCP allows hosts to obtain required TCP/IP configuration information from a DHCP server
  - Without DHCP, IP addresses for new computers or computers that are moved from one subnet to another must be configured manually
  - Entire process is automated and managed centrally

SERVER ROLES

- ***Dynamic Host Configuration Protocol (DHCP)***
- ***Benefits of DHCP***
  - **Reliable IP address configuration**
    - Minimizes configuration errors caused by manual IP address configuration
    - Prevents duplicate IP's on the network
  - **Reduced network administration**
    - Centralized and automated TCP/IP configuration
    - The ability to define TCP/IP configurations from a central location
    - ***Unused IP's are automatically reclaimed by server***

SERVER ROLES

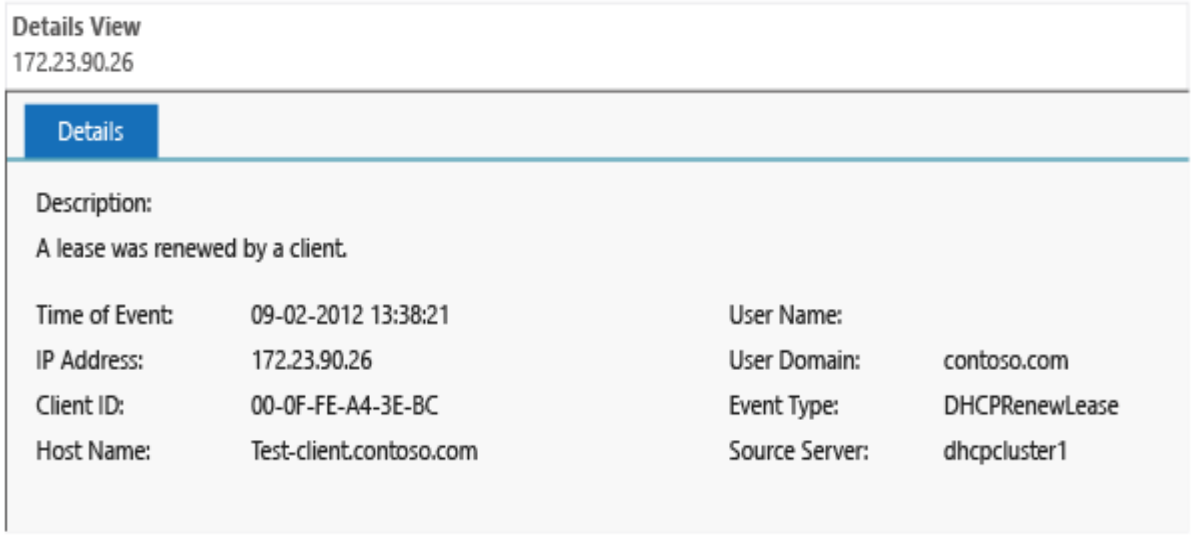
- ***Print and Document Services***
  - Enables the ability to centralize print server and network printer tasks
  - Can also receive scanned documents from network scanners and route the documents to a shared network resource
  - ***Ability to upload “printer drivers” to one location***

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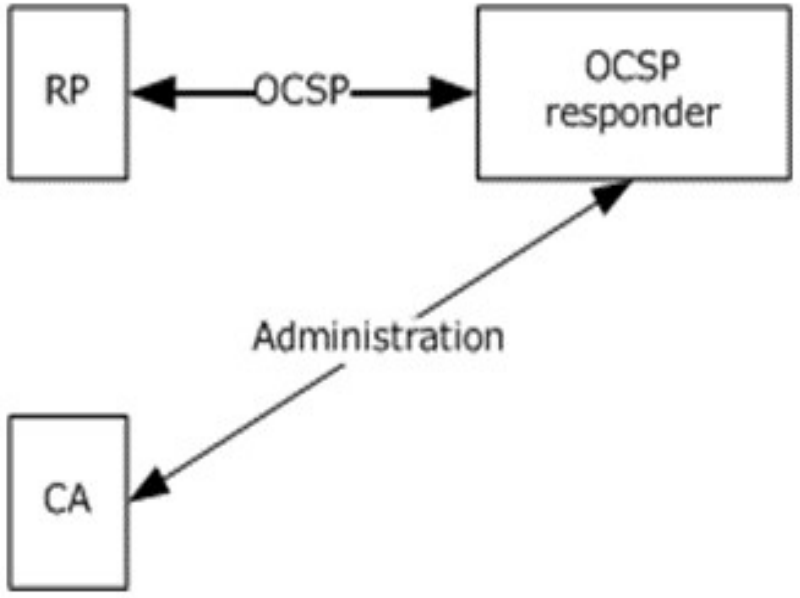
SERVER ROLES

- ***IP Address Management (IPAM)***
  - IPAM automatically discovers IP address infrastructure servers and Domain Name System (DNS) servers on the network
  - Enables administrators to manage them from a central interface



SERVER ROLES

- ***Microsoft Online Certificate Status Protocol (OCSP)***
  - Provides an efficient mechanism for distributing certificate revocation information
  - OCSP is a component of a public key infrastructure (PKI)



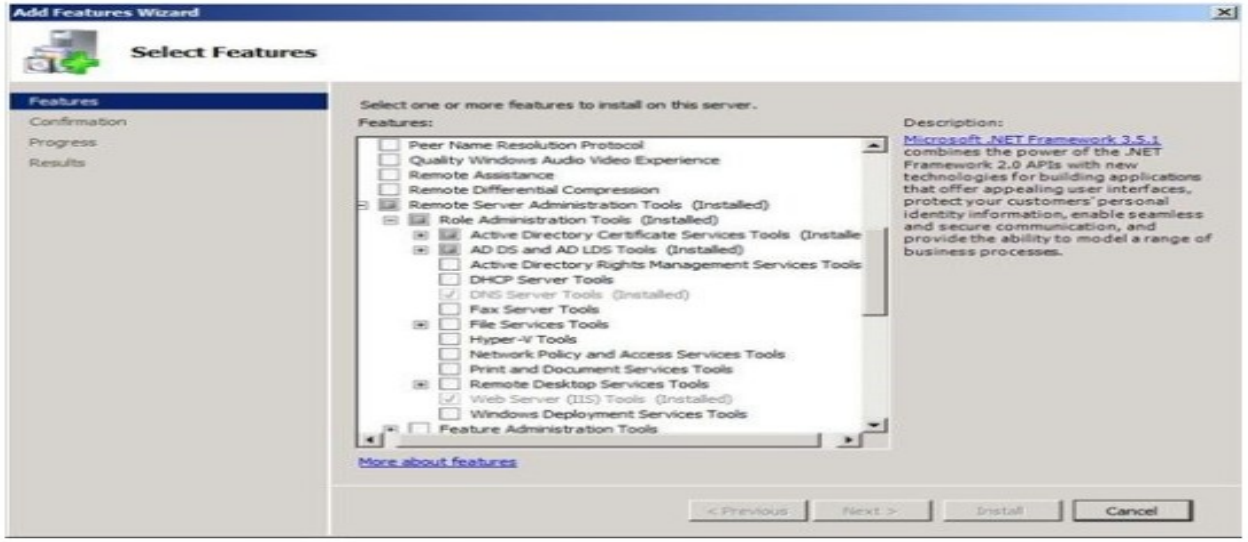
SERVER ROLES

- ***File and Storage Services***
  - Provide central locations on your network where you can store files and share them with users
  - ***How the “Shared Drive” is built***
  - Can be used for centralized backup and file management of documents

REMOTE SERVER ADMINISTRATIVE TOOLS (RSAT)

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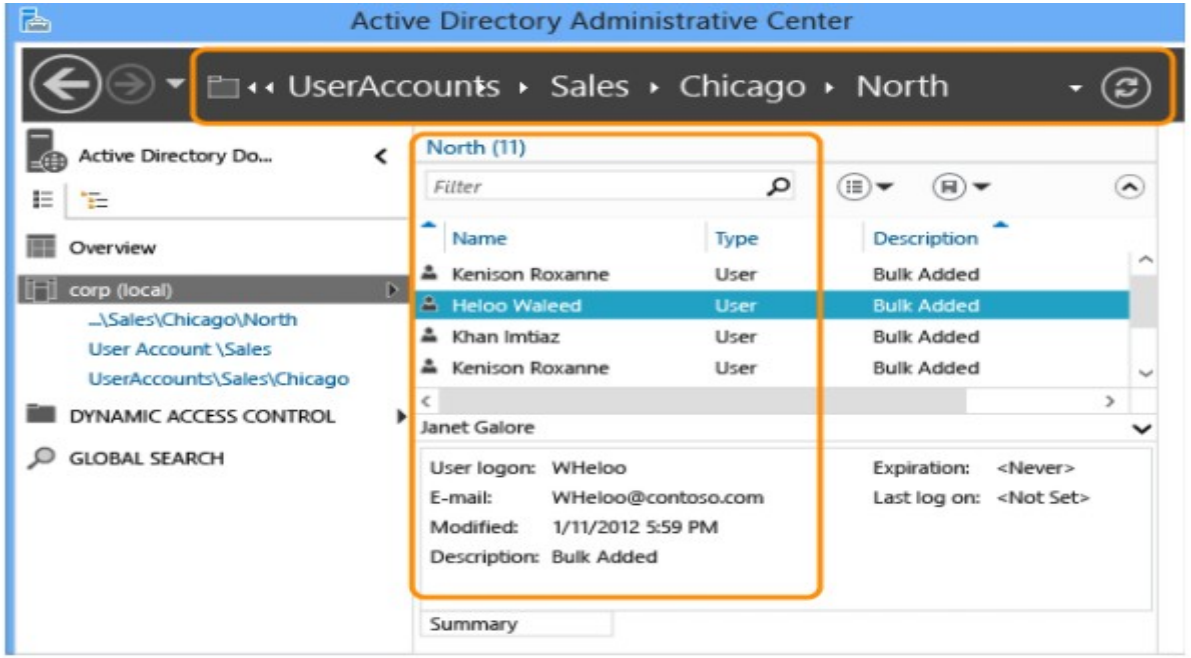
- ***Remote Server Administrative Tools (RSAT)***
  - Lets IT admins manage Windows Server roles and features from a Windows Workstation
  - Used to manage roles and features that are running on Remote servers
  - RSAT Tools includes Server Manager, Microsoft Management Console (mmc) snap-ins, consoles, Windows PowerShell cmdlets and providers, and some command-line tools



ACTIVE DIRECTORY ADMINISTRATION CENTER (ADAC)

ACTIVE DIRECTORY ADMINISTRATION CENTER (ADAC)

- ***Active Directory Administration Center (ADAC)***
  - Can do everything that ADUC can do but with more capabilities and the ability to have more control over individuals and groups
  - ADAC is the newer, primary tool for performing day to day tasks in the administration of the Active Directory environment



METADATA CLEANUP

ACTIVE DIRECTORY ADMINISTRATION CENTER (ADAC)

- ***Metadata cleanup***
  - Required procedure after a forced removal of a domain controller
  - Metadata cleanup removes data from ADDS that identifies a domain controller to the replication system
  - ***Multiple Steps needed to be done for proper removal***

REPADMIN TOOL

REPADMIN TOOL

- ***REPADMIN TOOL***
  - Tool called Repadmin that administrators can use to diagnose (and in some cases repair) Active Directory replication
  - Administrators must open an elevated Command Prompt window or PowerShell
  - *Refer to student handout for different commands

SUMMARY

- Server Roles
- RSAT
- ADAC
- Metadata Cleanup
- REPADMIN Tools


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